

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

06

10 answers · Rationale-rich · Teach from doc

Q1

4

The correct answer is 4. The equation $60h + 10 \leq 280$, where h is the number of hours the boat has been rented, can be written to represent the situation. Subtracting 10 from both sides and then dividing by 60 yields $h \leq 4.5$. Since the boat can be rented only for whole numbers of hours, the maximum number of hours for which Maria can rent the boat is 4.

Q2

C

C. $1.53a + 1.44b = 35(1.49)$

Choice C is correct. It's given that in 2010, a swim club had a total of 35 swimmers, each classified as either advanced or intermediate, and that a represents the number of advanced swimmers in 2010 and b represents the number of intermediate swimmers in 2010. It's also given that from 2010 to 2020, the number of advanced swimmers in the club increased by approximately 53% and the number of intermediate swimmers in the club increased by approximately 44%. Thus, in 2020, the approximate number of advanced swimmers in the club can be represented as $1.53a$ and the approximate number of intermediate swimmers in the club can be represented as $1.44b$. It's given that the total number of swimmers in the club increased by approximately 49% from 2010 to 2020. Since the club had 35 swimmers in 2010, it follows that the total number of swimmers in 2020 can be represented as $35(1.49)$. Since the sum of the number of advanced swimmers in 2020 and the number of intermediate swimmers in 2020 equals the total number of swimmers in 2020, the equation $1.53a + 1.44b = 35(1.49)$ best represents this situation.

Choice A is incorrect. This equation represents a situation where the number of intermediate swimmers in the club in 2020 increased by approximately 49%, not 44%, and the total number of swimmers in the club in 2020 increased by approximately 44%, not 49%.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. This equation represents a situation where the number of advanced swimmers in the club in 2020 increased by approximately 44%, not 53%, and the number of intermediate swimmers in the club in 2020 increased by approximately 53%, not 44%.

Q3**17**

The correct answer is 17. It's given that $2x + 3 = 9$. Multiplying each side of this equation by 3 yields $3(2x + 3) = 3(9)$, or $6x + 9 = 27$. Subtracting 9 from each side of this equation yields $6x + 9 - 9 = 27 - 9$, or $6x = 18$. Dividing each side by 6 yields $6x \div 6 = 18 \div 6$, or $x = 3$. Therefore, the value of x is 3.

Q4**B**

B. 250

Choice B is correct. Substituting 50 for x in the given function yields $h(50) = 50 + 200$, or $h(50) = 250$. Therefore, the value of $h(50)$ is 250.

Choice A is incorrect. This is the value of $h(0)$.

Choice C is incorrect. This is the value of $h(9,800)$.

Choice D is incorrect. This is the value of $h(50,000)$.

Q5**A**

A. (3,9)

Choice A is correct. The solution to this system of linear equations is the point that lies on both lines graphed, or the point of intersection of the two lines. According to the graphs, the point of intersection occurs when $x = 3$ and $y = 9$, or at the point (3,9).

Choices B and D are incorrect. Each of these points lies on one line, but not on both lines in the xy -plane. Choice C is incorrect. This point doesn't lie on either of the lines graphed in the xy -plane.

Q6**D**

D. $25 + 10t \leq 75$

Choice D is correct. The cost of the rental fee depends on the number of hours the surfboard is rented. Multiplying t hours by 10 dollars per hour yields a rental fee of $10t$ dollars. The total cost of the rental consists of the rental fee plus the 25 dollar service fee, which yields a total cost of $25 + 10t$ dollars. Since the person intends to spend a maximum of 75 dollars to rent the surfboard, the total cost must be at most 75 dollars. Therefore, the inequality $25 + 10t \leq 75$ represents this situation.

Choice A is incorrect. This represents a situation where the rental fee, not the total cost, is at most 75 dollars.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q7**A**

A. 0,81

Choice A is correct. In the xy -plane, the graph of an equation in the form $y = mx + b$, where m and b are constants, has a slope of m and a y -intercept of $(0, b)$. Therefore, the y -intercept of the graph of $y = 34x + 81$ is $(0, 81)$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q8**A****A.** $2c$

Choice A is correct. If one pound of grapes costs \$2, two pounds of grapes will cost 2 times \$2, three pounds of grapes will cost 3 times \$2, and so on. Therefore, c pounds of grapes will cost c times \$2, which is $2c$ dollars.

Choice B is incorrect and may result from incorrectly adding instead of multiplying. Choice C is incorrect and may result from assuming that c pounds cost \$2, and then finding the cost per pound. Choice D is incorrect and could result from incorrectly assuming that 2 pounds cost \$ c , and then finding the cost per pound.

Q9**20**

The correct answer is 20. For the graph shown, the x -axis represents the time since the initial deposit, in months, and the y -axis represents the bank account balance, in dollars. The amount of the initial deposit is estimated by the y -coordinate of the point on the graph that represents 0 months since the initial deposit. Therefore, the amount of the initial deposit is estimated by the corresponding y -value for the point when $x = 0$. When $x = 0$, it is estimated that $y = 20$. Thus, the amount of the initial deposit estimated by the graph, to the nearest whole dollar, is 20.

Q10

A

A. 4, - 5

Choice A is correct. The solution to this system of linear equations is represented by the point that lies on both lines shown, or the point of intersection of the two lines. According to the graph, the point of intersection occurs when $x = 4$ and $y = -5$, or at the point 4, - 5. Therefore, the solution x, y to the system is 4, - 5.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

Generated 2026-05-29 · cb-educator-question-bank